✏ Unit 6 Practice — Functions & Linear Relationships

Functions · Slope · Tables → Equations · Real-World · Comparing · Qualitative Graphs · Systems

A. Functions

Problem 1

Does the table shown represent a function? (yes/no)

x	y
1	5
2	7
3	9
4	11

yes

Problem 2

Does the table shown represent a function? (yes/no)

x	y
2	4
3	6
2	9
5	8

no

Problem 3

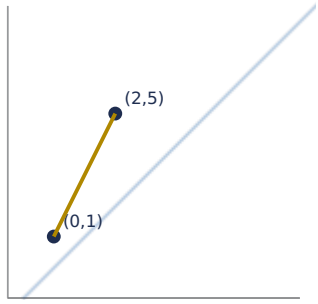
A relation includes the points (1,3), (2,5), (3,7), (1,9). Is it a function?

no

B. Slope

Problem 1

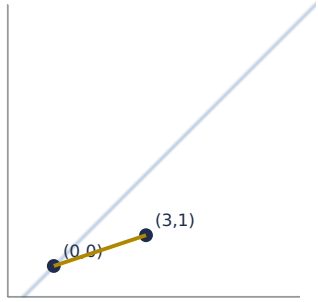
Find the slope of the line through the two points shown.



2

Problem 2

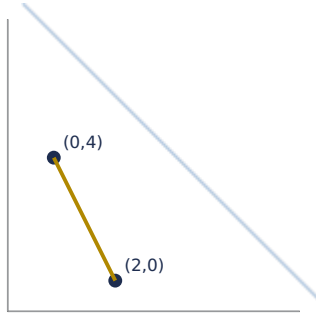
Find the slope of the line through the two points shown.



$\frac{1}{3}$

Problem 3

Find the slope of the line through the two points shown.



-2

C. Tables → Equations

Problem 1

Write the equation (y = mx + b) for the table shown.

x	y
0	1
1	3
2	5
3	7

y=2x+1

Problem 2

Write the equation (y = mx + b) for the table shown.

x	y
0	10
1	7
2	4
3	1

y=-3x+10

D. Real-World Applications

Problem 1

A taxi charges a $4 flat fee plus $2 per mile. Write the cost equation (y = mx + b) where x = miles.

y=2x+4

Problem 2

A gym charges $30 to join plus $25 per month. How much for 6 months total?

180

E. Comparing Functions

Problem 1

Function A: y = 3x + 2. Function B has a table showing slope 5. Which has the greater rate of change?

function b

Problem 2

Function A: y = 4x − 1. Function B passes through (0,3) and (2,7). Which has the greater y-intercept?

function b

F. Qualitative Graphs

Problem 1

A graph shows temperature rising steadily, then staying flat for an hour, then rising again. During the flat part, is temperature increasing, decreasing, or constant?

constant

Problem 2

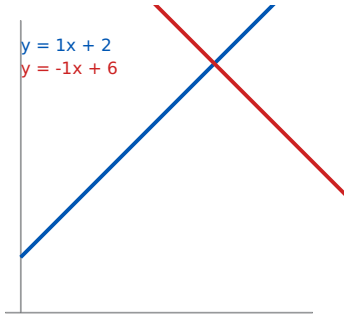
A car's distance-time graph shows a steep upward line, then a horizontal line. What is happening during the horizontal part?

stopped

G. Systems of Equations

Problem 1

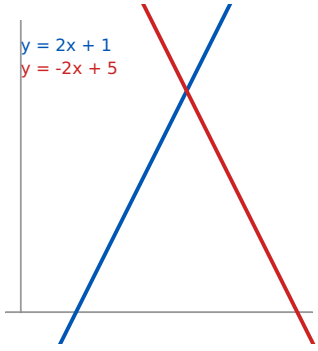
Find the solution (x,y) to the system of equations graphed.



(2,4)

Problem 2

Find the solution (x,y) to the system of equations graphed.



(1,3)

Problem 3

Solve the system: $y = x + 2$ and $y = 3x - 2$. Find x.

2